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Paleomagnetism of Quaternary Volcanics in the Izu Peninsula and Adjacent Areas, Japan, and Its Tectonic Significance

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A paleomagnetic investigation was made on the Quaternary volcanics in the Izu Peninsula and the adjacent Ashitaka and Izu-Oshima areas located on the northern tip of the Philippine Sea plate, central Japan. This study evaluates the deformation associated with the collision of the Izu block with Honshu. About 900 volcanic rock samples ranging in age from 0.2 to 1.5 Ma were measured. Alternating field demagnetization indicates that these samples have stable natural remanent magnetizations. Fine grained titanomagnetite was identified as the carrier of the remanent magnetizations by thermomagnetic analysis and microscopic observation. The paleomagnetic directions from both Ashitaka Volcano and Izu-Oshima island, which are located to the north and to the east of the Izu Peninsula, respectively, are not significantly different from that of the present axial geocentric dipole field. In contrast, the mean directions from the northwestern Izu Peninsula show on average 10° clockwise deflections, whereas the directions from the northeastern to southern Izu Peninsula show 8° to 25° counterclockwise deflections. These deflections of paleomagnetic directions are probably caused by tectonic deformations since 1.5 Ma. We propose a simple tilting model to explain these paleomagnetic directions.

1. Introduction

It is widely accepted that the Izu block, which is located on the northern tip of the Philippine Sea plate, is colliding with the Honshu arc (e.g., MATSUDA, 1978; NIITSUMA and MATSUDA, 1985; Fig. 1). On the basis of paleomagnetic results from the Izu Peninsula, HIROOKA *et al.* (1983, 1985) and KOYAMA (1983) noted that the Izu block originated at a latitude significantly lower than that of the present and has drifted northward until it reached the present position. In the light of geological and paleontological data around the Izu block, HUCHON and KITAZATO (1984) and KITAZATO (1986) suggested that the Izu block began to collide with Honshu between 0.8 and 1.5 Ma. All these results suggest that the Izu block has been subject to some deformation associated with the collision since the early Quaternary.

We usually evaluate tectonic deformation by studying geologic structure. In a volcanic sequence it is, however, generally difficult to clarify the geologic structure because of the lack of records of the initial horizontal surface (e.g., bedding plane of

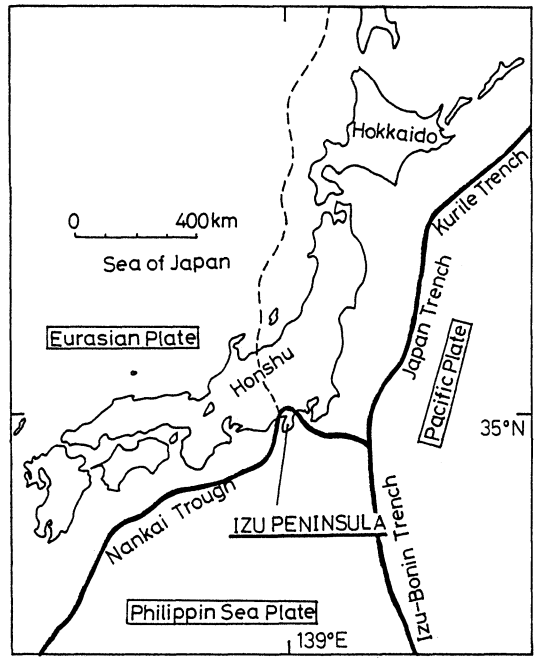


Fig. 1. Index map showing the study area. Thick solid line: plate boundary. Broken line: estimated plate boundary between the Eurasian and the North American plates (NAKAMURA, 1983) since 0.5 Ma (SENO, 1985).

subaqueous fine sediments). The Quaternary strata in the Izu Peninsula and Izu-Oshima island are mostly composed of volcanic rocks. Their distribution proves that they lie nearly horizontally or have gentle dips. The true sense and amount of tectonic tilting or rotation, which are keys to revealing the Quaternary deformation of the Izu block, are, however, hardly decided from field evidence. In this study, in situ paleomagnetic directions are used to estimate the true sense and amount of tectonic deformation recorded in volcanic rocks.

Several paleomagnetic studies were made on the Quaternary volcanics mainly in the northeastern to middle eastern Izu Peninsula and revealed that counter-clockwise deflections of declination up to several tens of degrees commonly exist in this area, such as in Taga Volcano (NAGATA *et al.*, 1957), Usami Volcano (KONO, 1968; KOYAMA, 1981), Tenshi and Amagi Volcanoes (KOYAMA, 1986). The declination deflection in the Usami Volcano was explained by local deformations along a large strike-slip fault (KOYAMA, 1981). However, the tectonic implication of the deflections in the other volcanoes is unclear yet. We paleomagnetically examine six volcanoes in the Izu Peninsula, one to the north of it, and three on Izu-Oshima island to the east of the Izu Peninsula (Fig. 2), and try to detect the Quaternary deformation, which is expected to be associated with the collision of the Izu block

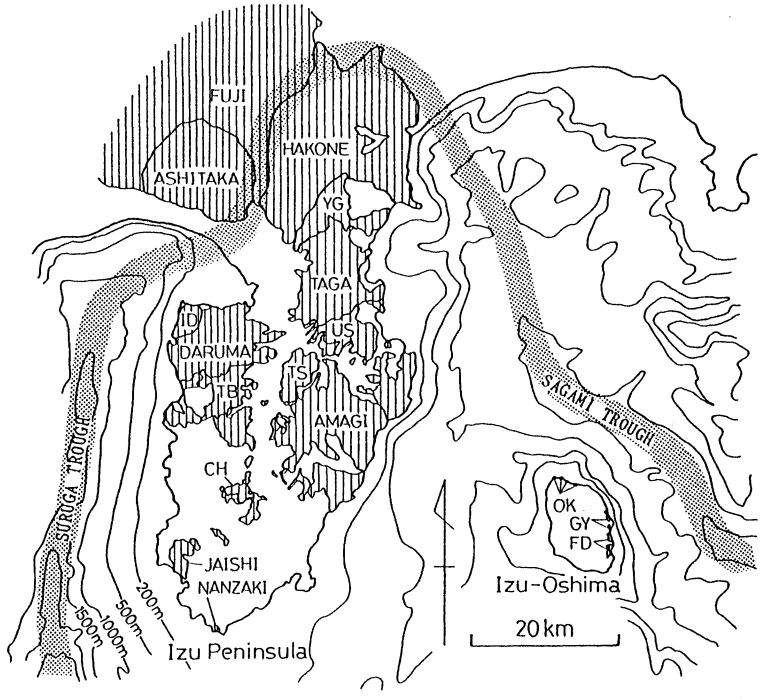


Fig. 2. Distribution of the Quaternary volcanics in the Izu Peninsula and adjacent areas. Names of major Quaternary volcanoes are also shown: ID, Ida Volcano; TB, Tanaba Volcano; CH, Chokuro Volcano; TS, Tenshi Volcano; US, Usami Volcano; YG, Yugawara Volcano; OK, Okata Volcano; GY, Gyojanoiwaya Volcano; and FD, Fudeshima Volcano. Shaded belt: material boundary between the Philippine Sea plate and Honshu (NAKAMURA *et al.*, 1984).

against Honshu. Previous paleomagnetic studies on the volcanoes in the northeastern to middle eastern Izu Peninsula are referred to as well.

2. Geology and Paleomagnetic Samples

Quaternary volcanics in and around the Izu Peninsula are products of several terrestrial volcanoes, which are composed of andesitic, basaltic, and a small amount of dacitic lavas and pyroclastics (e.g., OKI *et al.*, 1978). The present paleomagnetic measurements were made on the following ten volcanoes: Ida, Daruma, Tanaba, Chokuro, Jaishi, and Nanzaki Volcanoes in the Izu Peninsula; Ashitaka Volcano to the north of the Izu Peninsula; Fudeshima, Okata, and Gyojanoiwaya Volcanoes on Izu-Oshima island (Fig. 2).

The paleomagnetic samples were collected at 135 sites (Table 1). Most of the samples were taken from non-brecciated parts of lava flows, except for those of Fudeshima Volcano which were sampled from dykes. Each sampling site was

Table 1. Summary of paleomagnetic results for each sampling site. N is number of specimens; ODF is intensity of optimum demagnetizing field (see text); INCL and DECL are inclination and declination of remanent magnetization after alternating field demagnetization; k is precision parameter; A_{95} is radius of 95% confidence limit in degrees; VGPLAT and VGPLONG are latitude and longitude of virtual geomagnetic pole (VGP); K is initial magnetic susceptibility ($\times 10^{-3}$ G/Oe); J_n is intensity of natural remanent magnetization ($\times 10^{-3}$ emu/g); Q_n is Koenigsberger ratio; MDF is intensity of median demagnetizing field (Oe); T_{C1} and T_{C2} are Curie temperature in heating and cooling stages (°C); J_o is saturation magnetization before heating (emu/g); J_h/J_o is ratio of saturation magnetization after thermal treatment to J_o; Type is a type of thermomagnetic curves (see text).

Site	N	ODF	INCL	DECL	k	A_{95}	VGP-LAT	VGP-LONG	K	J_n	Q_n	MDF	T_{C1}	T_{C2}	J_o	J_h/J_o	Type
Ashitaka Volcano																	
AS01	5	NRM	30.4	16.1	175	5.8	66.3N	277.4E	3.4	0.5	0.8	225	530	520	3.2	1.0	I
02	4	NRM	46.5	28.7	251	5.8	64.5N	237.7E	2.1	0.3	0.7	225					
03	6	100	13.5	355.8	103	6.6	61.4N	327.5E	2.8	0.4	0.8	175	560	540	2.2	1.0	I
04	6	150	24.7	16.3	50	9.6	63.3N	281.3E	3.1	0.4	0.7	225					
05	7	NRM	49.6	334.7	111	5.8	68.3N	43.3E	3.4	0.9	1.4	250	510	485	3.5	0.9	I
06	7	NRM	36.9	354.8	40	9.7	74.7N	337.5E	3.3	0.4	0.6	200	540	530	1.7	1.2	II
07	5	100	39.0	344.9	40	12.3	71.4N	367.9E	3.5	0.5	0.7	100					
08	7	NRM	51.2	334.4	162	4.8	68.5N	47.3E	3.3	0.7	1.3	150	520	480	2.4	1.0	II
09	6	NRM	45.8	358.2	253	4.2	81.9N	330.2E	3.0	0.9	1.6	175					
10	8	NRM	59.3	342.3	60	7.2	75.2N	73.4E	3.2	0.6	1.0	175	545	530	3.0	1.0	I
11	6	NRM	54.3	328.4	29	12.7	64.2N	57.3E	2.9	0.7	1.3	150					
12	7	NRM	47.6	339.1	264	3.7	71.2N	34.5E	3.1	0.6	1.1	150	530	510	2.0	1.0	I
13	6	200	62.4	331.9	33	11.9	66.8N	78.9E	2.2	0.3	0.7	250	570	535	0.3	0.9	I
14	7	NRM	53.5	319.7	123	8.3	57.0N	58.7E	3.1	0.5	0.9	175					
15	7	NRM	56.8	347.9	222	4.1	80.0N	65.0E	3.0	0.8	1.5	175	510	480	2.5	1.0	I
16	7	150	56.3	22.2	49	8.7	72.0N	217.1E	2.1	1.7	4.8	200					
17	5	NRM	58.1	43.1	106	7.5	55.7N	209.8E	3.4	1.7	2.7	275	490	470	3.0	0.9	I
18	7	200	56.5	351.5	43	9.3	82.9N	66.3E	3.5	1.2	2.3	275					
19	6	NRM	46.2	13.2	499	3.0	76.4N	259.6E	3.0	1.8	3.0	300					
Izu-Oshima Island																	
Okata Volcano																	
OS01	5	300	63.9	346.4	50	10.9	75.1N	99.7E	2.7	4.1	6.3	200	530	520	2.2	0.9	I
02	6	300	59.0	341.2	68	8.2	74.3N	73.3E	2.1	3.5	6.9	350					
03	5	NRM	57.5	359.4	538	3.3	86.6N	131.3E	2.9	2.4	5.8	225	510	500	2.6	0.9	I
04	4	100	26.3	2.7	86	9.9	68.9N	312.1E	2.3	1.3	2.6	80	500	460	1.6	1.0	I
05	5	NRM	51.3	346.5	407	3.8	78.4N	39.1E	2.8	4.6	11.2	250					
22	5	NRM	32.5	29.4	302	4.4	58.8N	255.0E	1.9	1.3	4.0	300	500	485	0.8	0.9	I
23	5	100	21.1	17.0	146	6.4	61.6N	282.3E	2.2	1.5	4.1	125					
Gyojanoiwaya Volcano																	
OS06	12	50	47.7	350.1	38	7.5	80.3N	22.2E	2.2	0.9	3.0	100	515	480	1.8	1.0	I

Table 1. (continued).

Site	N	ODF	INCL	DECL	k	A_{95}	VGP-LAT	VGP-LONG	K	J_n	Q_n	MDF	T_{C1}	T_{C2}	J_o	J_h/J_o	Type
Fudeshima Volcano																	
OS07	3	NRM	76.5	298.6	125	11.1	43.1N	108.6E	2.1	0.6	1.7	350	540	530	1.1	1.0	III
08	4	NRM	54.7	354.2	40	14.8	85.2N	57.4E	2.4	1.9	3.5	175					
09	3	NRM	65.1	304.9	750	4.5	47.5N	83.8E	2.6	0.4	1.1	175	560	530	0.6	0.6	I
10	4	75	58.8	356.9	77	10.5	84.6N	113.1E	2.3	2.3	5.1	375	530	480	1.3	0.7	I
11	5	NRM	57.8	9.0	503	3.4	81.9N	199.4E	1.8	0.3	0.9	225					
12	5	100	42.6	27.6	47	11.3	64.1N	244.7E	2.4	1.0	2.3	225					
13	4	NRM	58.0	37.4	86	9.9	60.0N	210.8E	2.0	1.0	3.1	250	490	460	2.1	1.1	II
14	4	NRM	40.7	11.0	230	6.1	75.1N	276.6E	2.0	1.1	3.7	250	560	520	1.2	0.8	I
15	5	NRM	58.4	357.7	160	6.1	85.2N	117.4E	2.0	0.9	2.5	200					
16	4	NRM	50.0	356.9	399	4.6	85.3N	354.0E	2.8	0.9	1.7	175	550	520	1.0	0.9	I
17	4	NRM	52.4	351.0	469	4.3	82.3N	39.1E	4.6	1.7	2.7	225					
18	4	NRM	43.9	350.8	697	3.5	78.0N	3.3E	2.4	5.8	12.6	375					
19	5	NRM	56.4	346.9	143	6.4	79.2N	65.3E	3.5	7.9	13.0	300					
20	4	NRM	52.5	345.9	273	5.6	78.2N	45.6E	2.6	5.9	14.7	475	560	540	2.4	0.8	I
21	4	100	55.3	356.2	81	10.3	86.7N	70.6E	2.9	3.1	6.3	225					
24	5	100	39.9	16.0	84	8.4	71.6N	265.9E	2.3	1.5	4.2	200	480	440	1.5	1.2	II
25	7	NRM	48.7	8.8	203	4.2	81.0N	261.3E	2.1	0.7	2.2	275					
26	5	NRM	43.5	358.9	825	2.7	80.6N	325.6E	2.3	2.5	6.8	150					
27	5	100	25.2	24.6	113	7.3	59.1N	267.4E	1.9	1.8	5.6	200	320	340	1.1	1.1	III
28	4	NRM	30.3	5.7	289	5.4	70.9N	302.5E	2.3	2.0	6.4	375					
29	4	NRM	46.8	7.4	242	5.9	80.8N	274.1E	2.0	1.1	3.3	250	520	470	1.3	0.9	I
Izu Peninsula																	
Ida Volcano																	
ID02	9	100	54.7	356.1	57	6.9	86.8N	54.2E	2.4	0.6	1.6	125					
03	8	150	29.8	359.2	28	10.6	71.0N	321.2E	2.6	0.6	1.5	250					
04	3	100	60.3	6.8	98	10.2	81.8N	177.3E	2.4	0.9	2.1	250	575	555	2.4	0.9	I
11	5	100	63.2	26.2	144	6.4	67.8N	195.0E	2.6	0.4	1.0	140	585	560	2.1	0.7	I
21	4	150	45.4	4.4	66	11.4	81.0N	292.8E	2.8	1.1	2.2	110	540	530	2.8	1.1	II
27	19	150	51.3	1.7	58	8.9	86.7N	293.1E	3.0	0.6	1.2	160					
44	5	NRM	58.6	24.8	410	3.2	69.8N	209.0E	3.5	1.8	3.1	225					
45	5	NRM	55.7	31.2	1231	2.2	64.7N	216.8E	3.2	1.7	3.2	110	530	520	2.5	1.0	I
56	6	150	29.3	15.9	143	5.6	66.0N	278.4E	2.5	1.0	2.5	225					
57	5	NRM	11.0	12.4	283	4.6	58.4N	294.7E	2.5	0.9	2.3	80	525	510	2.4	0.9	I

Table 1. (continued).

Site	N	ODF	INCL	DECL	k	A_{95}	VGP-LAT	VGP-LONG	K	J_n	Q_n	MDF	T_{C1}	T_{C2}	J_o	J_h/J_o	Type
Daruma Volcano: late lava flows of normal polarity																	
DR01	4	NRM	45.0	24.8	405	4.6	67.2N	243.5E	2.8	1.4	2.7	275					
02	5	NRM	36.1	333.4	89	8.2	62.2N	23.3E	2.5	0.2	0.5	125					
03	14	NRM	43.3	1.9	282	3.3	80.1N	308.7E	3.2	0.7	1.4	120					
04	5	NRM	51.7	354.1	1290	2.1	84.4N	22.3E	2.9	1.4	2.8	130					
05	6	NRM	49.6	321.8	101	6.7	57.7N	51.7E	3.4	0.2	0.4	150					
06	5	NRM	14.7	18.1	84	8.4	57.9N	283.4E	2.9	0.2	0.3	130					
16	6	NRM	46.3	22.5	348	3.6	69.5N	243.5E	2.6	0.6	1.4	130					
17	9	200	48.3	29.7	163	5.3	64.3N	233.2E	3.2	0.5	0.9						
19	12	150	37.9	7.2	88	4.7	74.9N	292.2E	2.9	0.6	1.5	140					
22	25	100	20.6	347.2	12	12.3	63.0N	347.5E	3.1	0.3	0.3	80					
25	4	NRM	64.0	349.7	250	5.8	76.7N	106.0E	3.0	3.0	4.4	350					
29	6	75	54.3	31.7	10	22.4	64.1N	219.9E	3.2	0.2	0.4	175					
30	6	NRM	54.0	16.3	391	3.4	76.6N	225.8E	2.7	2.0	4.1	225					
31	6	NRM	50.7	22.2	280	4.0	71.1N	233.2E	2.9	1.1	2.3	175					
32	6	NRM	50.8	4.2	633	2.7	85.1N	271.9E	2.9	1.6	3.4	225					
33	8	NRM	45.6	356.7	350	2.8	81.6N	339.4E	2.5	0.7	1.8	130	520	485	2.4	0.9	I
34	6	100	54.0	25.2	79	8.7	69.3N	222.6E	2.9	0.4	1.0	90					
35	4	NRM	63.2	24.6	287	5.4	68.8N	193.8E	2.3	2.1	5.1	275					
36	6	NRM	60.2	27.3	1107	2.0	67.7N	204.5E	1.6	3.0	10.5	300	535	525	1.0	1.0	I
37	5	NRM	56.5	28.1	281	4.6	67.3N	215.1E	2.0	1.5	4.4	275					
54	6	NRM	48.7	11.0	304	3.9	79.3N	255.8E	2.2	0.9	2.4	85	540	535	2.0	1.1	II
Daruma Volcano: late lava flows of reversed polarity																	
DR07	6	100	−43.0	225.0	32	12.1	50.1S	50.6E	3.1	0.3	0.7	225	520	495	2.3	1.0	I
08	4	NRM	−46.6	189.2	1773	1.8	79.4S	88.4E	1.2	1.6	7.1	800	585	575	0.2	2.0	I
09	6	NRM	−63.6	226.7	58	8.9	53.4S	18.0E	2.5	0.1	0.3	400					
10	8	NRM	−55.8	202.7	225	3.2	71.6S	37.9E	3.2	1.1	3.2	180	550	540	3.3	0.9	I
11	15	150	−48.7	185.4	169	3.5	83.0S	96.6E	3.1	1.3	2.6	475	525	440	3.0	0.9	I

Table 1. (continued).

Site	N	ODF	INCL	DECL	k	A_{95}	VGP-LAT	VGP-LONG	K	J_n	Q_n	MDF	T_{C1}	T_{C2}	J_o	J_h/J_o	Type
DR12	6	NRM	−60.9	175.5	487	3.0	82.2S	93.4E	2.2	1.2	3.1	300					
18	20	150	−46.2	178.8	51	9.4	82.5S	47.0E	3.3	0.5	0.9	50					
20	6	100	−80.6	167.5	43	11.5	52.7S	12.4E	2.7	0.5	0.9	150					
21	6	NRM	−34.7	166.9	317	3.8	70.4S	78.4E	2.7	0.6	1.2	225	515	490	2.1	1.0	I
23	12	100	−62.7	160.7	226	2.9	72.6S	66.3E	2.5	1.6	3.5	125					
24	21	100	−45.6	186.5	13	12.4	80.3S	1.9E	2.6	0.4	0.9	140	525	495	2.3	1.0	II
26	9	150	−15.3	196.7	16	13.4	58.8S	5.5E	2.2	1.3	2.8	25					
28	16	NRM	−29.5	195.8	1161	2.7	66.2S	98.4E	2.4	0.5	1.2	600	545	530	0.7	1.0	I
40	5	NRM	−48.5	205.2	405	3.8	68.1S	56.2E	2.8	0.3	0.7	160					
41	6	NRM	−49.2	181.3	164	5.3	85.0S	25.8E	2.8	0.6	1.3	150					
42	4	NRM	−49.1	197.1	333	4.2	74.8S	63.1E	1.7	1.8	6.0	375	555	530	1.5	0.9	I
43	5	NRM	−54.7	189.7	203	5.4	66.2S	98.4E	2.3	0.6	1.5	75	540	535	2.4	1.0	I
55	6	NRM	−59.8	188.3	250	4.3	81.3S	5.2E	2.9	0.8	1.7	160					
DD01	9	150	−58.7	196.3	56	7.0	76.3S	25.1E	2.2	0.2	0.7	250	540	520	1.4	1.2	II
02	8	NRM	−64.1	196.7	135	4.8	73.3S	3.1E	2.2	0.2	0.5	400	555	550	1.5	1.1	II
Daruma Volcano: early lava flows of normal polarity																	
DR46	8	150	35.8	45.6	4	27.4	47.2N	237.2E	2.8	0.1	0.3	150					
49	4	NRM	55.4	36.1	147	7.6	60.7N	216.3E	4.2	0.9	1.2	160					
50	4	NRM	56.7	10.7	566	3.9	81.1N	210.7E	2.9	0.8	1.5	700	570	565	1.2	1.0	I
51	4	100	76.0	37.9	114	8.6	53.3N	166.1E	2.9	0.3	0.5	200					
Daruma Volcano: early lava flows of reversed polarity																	
DR47	11	600	−22.1	177.0	5	21.8	66.4S	146.2E	2.1	0.2	0.2	800					
48	4	NRM	−23.2	171.4	115	8.6	65.8S	159.7E	1.3	0.1	0.2	800					
Tanaba Volcano																	
TA01	6	75	−24.9	178.0	14	18.3	68.1S	144.1E	2.9	0.1	0.2	100	520	480	2.0	1.1	II
02	7	100	−30.5	189.2	26	12.2	69.8S	112.6E	2.8	0.4	0.6	125					
03	6	100	−56.6	218.9	44	10.2	58.7S	33.2E	2.4	0.2	0.5	200	550	535	1.6	1.2	II
04	8	NRM	−75.7	181.6	56	7.4	61.9S	320.4E	2.6	0.3	0.5	150	520	520	1.9	1.3	II
05	7	150	−51.2	172.7	123	5.5	83.3S	204.2E	3.2	0.4	0.6	150					
06	8	100	−49.3	210.4	47	8.2	64.0S	50.7E	3.4	0.3	0.7	150	540	530	3.0	1.0	III
07	6	NRM	−34.5	193.6	156	5.4	70.0S	98.2E	3.1	1.2	2.3	200					
08	5	NRM	−51.6	176.7	107	7.4	86.2S	185.9E	3.4	0.8	1.2	325	550	530	3.0	0.9	I

Table 1. (continued).

Site	N	ODF	INCL	DECL	k	A_{95}	VGP-LAT	VGP-LONG	K	J_n	Q_n	MDF	T_{C1}	T_{C2}	J_o	J_h/J_o	Type
Chokuro Volcano																	
CH01	8	NRM	39.2	347.5	59	7.9	73.3N	3.2E	2.9	0.4	0.7	75	550	520	1.7	1.0	II
02	8	NRM	45.1	324.0	238	3.9	58.2N	44.1E	2.8	0.5	1.0	150	555	545	2.4	1.0	III
03	5	100	49.1	7.2	13	21.9	82.3N	265.1E	3.0	0.4	0.8	100					
04	6	100	41.8	352.4	55	10.4	77.5N	352.6E	3.0	0.3	0.7	80					
05	7	NRM	42.5	353.5	91	6.3	78.4N	349.6E	2.9	0.4	0.7	100	560	545	1.6	1.0	III
06	6	300	17.7	7.7	151	6.2	63.3N	301.7E	3.0	0.4	0.8	125					
07	5	NRM	35.5	348.7	359	4.0	71.9N	355.2E	3.1	0.4	0.7	100	580	550	1.2	0.7	I
Jaishi Volcano																	
JA01	7	NRM	−53.2	145.3	1019	1.9	61.4S	237.0E	2.4	0.6	1.5	150	535	520	1.8	1.0	I
02	6	NRM	−22.7	117.1	86	7.3	28.9S	203.2E	3.0	0.3	0.6	125					
05	6	150	−65.0	161.0	56	16.6	71.2S	275.3E	3.9	24.0	29.0	130	500	440	2.8	1.0	I
06	7	NRM	−50.0	156.1	349	3.2	69.6S	224.4E	3.3	0.6	1.0	150					
07	7	NRM	−60.1	172.3	60	7.8	81.3S	277.1E	3.2	0.7	1.4	225	520	470	2.2	1.1	II
08	6	NRM	−57.1	187.0	151	5.5	83.6S	18.9E	3.3	0.5	0.9	325	540	510	2.8	0.9	I
10	7	NRM	−54.1	187.3	63	6.3	84.0S	47.2E	3.4	0.9	1.6	200					
11	7	500	−68.1	152.9	42	9.4	64.4S	277.3E	3.1	0.5	1.1	600	530	510	2.1	0.9	I
12	7	NRM	−69.2	132.9	158	4.8	52.3S	272.4E	4.0	0.9	1.3	375					
Nanzaki Volcano																	
MI01	6	NRM	32.2	352.0	115	6.3	71.4N	343.4E	3.6	1.1	1.7	200	480	470	3.2	0.7	I
02	7	150	32.6	352.9	80	6.8	72.0N	341.2E	3.5	4.0	7.4	60	510	480	2.3	0.9	I
03	6	200	40.4	355.4	58	8.9	77.8N	339.2E	3.5	1.1	2.2	250					
04	4	NRM	39.5	345.7	149	7.6	72.5N	8.3E	2.4	2.0	5.1	325	550	520	1.0	0.8	I
05	7	NRM	38.9	335.2	406	3.0	64.9N	25.2E	3.5	0.8	1.6	375					

selected to represent a different cooling unit. In each sampling site, three to twenty-five oriented block samples were collected. They are spaced at least one meter apart from each other within the site. All of the samples were oriented with a magnetic compass. The accuracy of orientation is estimated to be less than 4° on the basis of repeated orientation and comparison with measurements using a sun compass. The oriented block samples were drilled and cut into core specimens (25 mm in diameter and height) in the laboratory.

The general geology of each volcano, horizon, number, and lithology of paleomagnetic samples are summarized below.

Ashitaka Volcano: Ashitaka Volcano is a basaltic to andesitic stratovolcano situated to the north of the Izu Peninsula (YUI and FUJII, 1988) (Fig. 2). According to YUI and FUJII (1988), Ashitaka Volcano is divided into lower, middle and upper parts, and the age of the middle and upper parts ranges from 0.1 to 0.2 Ma in the light of tephrochronology. Paleomagnetic samples were collected from the lower part at 19 sites of lava flows (Table 1).

Fudeshima Volcano: Fudeshima Volcano is exposed along the southeastern coast of Izu-Oshima island (ISSHIKI, 1984) (Fig. 2). It consists of an alternation of basaltic lava flows and pyroclastics intruded by several tens of basaltic dykes. The older limit of K–Ar age of this volcano is 2.4 Ma (KANEOKA *et al.*, 1970). Paleomagnetic samples were collected at 21 dyke sites (Table 1).

Okata Volcano: Okata Volcano, which is exposed along the northern coast of Izu-Oshima island, consists of an alternation of basaltic to andesitic lava flows and pyroclastics (ISSHIKI, 1984) (Fig. 2). The older limit of K–Ar age of this volcano is 0.42 Ma (KANEOKA *et al.*, 1970). Paleomagnetic samples were collected at seven sites of basaltic lava flows (Table 1).

Gyojanoiwaya Volcano: Gyojanoiwaya Volcano is exposed along the eastern coast of Izu-Oshima island (ISSHIKI, 1984) (Fig. 2). It is composed of an alternation of basaltic lava flows and scoria. Paleomagnetic samples were collected at the site of one lava flow (Table 1).

Ida Volcano: Ida Volcano is a basaltic stratovolcano situated in the northwestern Izu Peninsula (SHIRAO, 1981) (Fig. 2). The K–Ar age of one lava sample was estimated to be 0.64 ± 0.05 Ma (KANEOKA *et al.*, 1988). Paleomagnetic samples were collected at ten sites of lava flows (Table 1).

Daruma Volcano: Daruma Volcano is an andesitic shield volcano located to the south of Ida Volcano (Fig. 2) and underlies a part of it (SHIRAO, 1981). According to SHIRAO (1981), Daruma Volcano is divided into three parts: early lava flows, late lava flows, and flank deposits. Six K–Ar ages of the late lava flows range from 0.59 to 0.83 Ma (KANEOKA *et al.*, 1988). Paleomagnetic samples were collected at six sites of early lava flows, and at 41 sites of late lava flows (Table 1).

Tanaba Volcano: Tanaba Volcano is an andesitic stratovolcano located to the south of Daruma Volcano (Fig. 2) and underlies a part of it (SHIRAO, 1981). Three K–Ar ages ranging from 1.19 to 1.52 Ma were reported (AGENCY OF NATURAL RESOURCES AND ENERGY (ANRE), 1987). Paleomagnetic samples were collected at eight sites of lava flows (Table 1).

Chokuro Volcano: Chokuro Volcano, which is situated in the middle western Izu Peninsula, is composed of several andesitic lava flows (SAWAMURA *et al.*, 1970) (Fig. 2). A K–Ar age of 0.62±0.07 Ma was reported for this volcano (ANRE, 1987). Paleomagnetic samples were collected at seven sites of lava flows (Table 1).

Jaishi Volcano: Jaishi Volcano is an andesitic stratovolcano located in the southern Izu Peninsula (CHIBA *et al.*, 1982) (Fig. 2). A K–Ar age of 1.35±0.07 Ma was reported for this volcano (ANRE, 1987). Paleomagnetic samples were collected at nine sites of lava flows (Table 1).

Nanzaki Volcano: Nanzaki Volcano (*1), which is situated in the southernmost Izu Peninsula, consists of a few basanitoid lava flows and a scoria cone (KURODA, 1976) (Fig.2). One of the lava flows gives a K–Ar age of 0.43±0.03 Ma (KANEOKA *et al.*, 1982). Paleomagnetic samples were collected at five sites of lava flows (Table 1).

(*1) KOYAMA and NIITSUMA (1980) used the name "Minamisaki" instead of "Nanzaki" because the name "Nanzaki" is not an official one for calling the area around the volcano. However, the name "Nanzaki" is currently used by the people in the area and has already been used in some papers. Thus, the present study uses the name "Nanzaki" to avoid confusion.

3. Measurement and Magnetic Properties

All the paleomagnetic samples were measured and demagnetized at the Ocean Research Institute, University of Tokyo. The remanent magnetizations of the samples were measured with a Schonstedt spinner magnetometer model SSM-1A. All the samples were subjected to alternating field (AF) demagnetization and the stability of remanence was examined. The demagnetizer consists of a uniaxial demagnetizing coil and a specimen tumbler revolving around two axes in a magnetic shield.

One to ten (mostly three to five) specimens from each site were stepwise demagnetized up to 800 Oe (80 mT) of peak field and the stability of remanence was tested. Other magnetic properties of the samples were examined through thermomagnetic analysis, magnetic susceptibility measurement, and microscopic observation of polished thin sections. Thermomagnetic analysis was made with an automatic torque balance in a magnetic field of 4500 Oe (450 mT) and with a vacuum of $10^{-5\sim-6}$ torr (approximately $10^{-3\sim-4}$ Pa). On the basis of the thermomagnetic curves, Curie temperatures were determined by the graphical method after GROMME *et al.* (1969). The magnetic susceptibility was measured with a Bison Instruments susceptibility meter model 3101A. In calculating Koenigsberger ratios, we used a value of 0.465 G (46.5 μT) as the intensity of the present geomagnetic field.

Magnetic properties of the present samples are summarized in Table 1, and a brief explanation for each parameter is given in the following.

Remanence intensity

NRM intensities (J_n) of the samples are classified into the following two groups. The samples of Ashitaka, Ida, Daruma, Tanaba, Chokuro, Jaishi, and Gyojanoiwaya

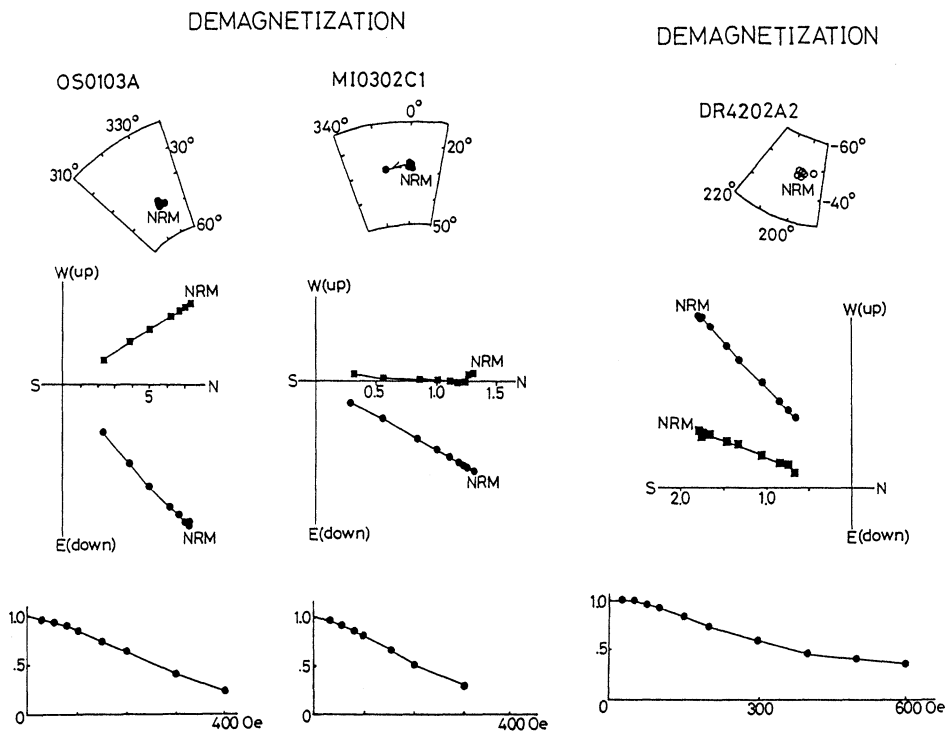


Fig. 3. Typical change of magnetization vectors against alternating field demagnetization. Upper figure shows directional changes of the vectors. Middle figure is a Zijderveld diagram of which scale unit is 1×10^{-3} emu/g ($=Am^2/kg$). Lower figure shows changes of normalized NRM intensity as a function of demagnetizing field intensity. The samples are of Okata Volcano (OS0103A), Nanzaki Volcano (MI0302C1), Chokuro Volcano (CH0202A), and Daruma Volcano (DR3001B2 and DR4202A2).

Volcanoes have lower intensities, most of which are 10^{-4} emu/g ($=Am^2/kg$) in the order of magnitude. Most of the samples of Nanzaki, Okata and Fudeshima volcanoes have higher intensities of 10^{-3} emu/g in the order of magnitude.

Stability against alternating field demagnetization

In many samples, NRM directions change little through stepwise AF demagnetizations (Fig. 3). This indicates that secondary components of remanence in the samples are generally small. The intensity of remanent magnetization decreases systematically with the increase in intensity of the demagnetizing field. The majority of median demagnetizing field intensities (MDF) are 50 to 300 Oe (5 to 30 mT).

Optimum demagnetizing field intensity (ODF) for each site was chosen as stated in the following. In a site with two or more stepwise demagnetized specimens, the intensity of demagnetizing field making minimum scatter of remanence directions was chosen as the ODF for the site (Table 1). Then the other specimens from the same site were demagnetized in the ODF and the site-mean direction after this

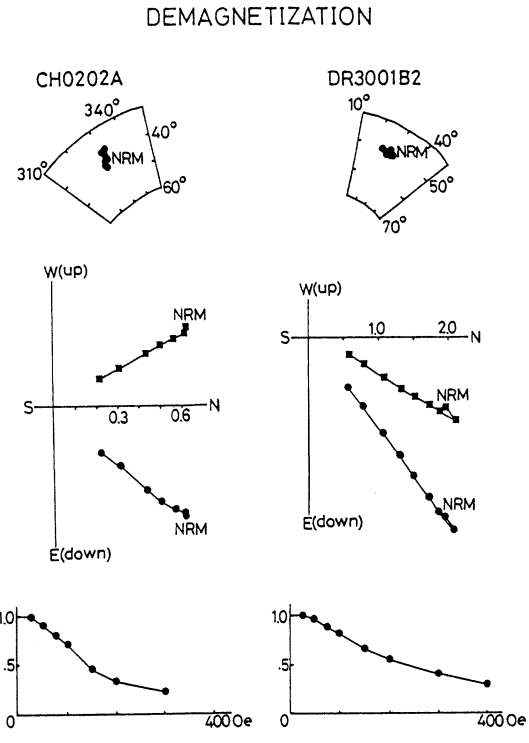


Fig. 3. (continued).

treatment was adopted as that of the stable component. In a site with one stepwise demagnetized specimen, a Zijderveld diagram (ZIJDERVELD, 1967) was used to deduce the ODF and the stable component direction.

Magnetic susceptibility

Most of the site-mean values of initial susceptibility (K) range from 2.0 to 3.5×10^{-3} G/Oe (1.6 to 2.8×10^{-4} in SI units). The samples of Okata and Fudeshima volcanoes have higher J_n than those of the other samples in spite of the relatively lower K values of 2.4×10^{-3} G/Oe (1.9×10^{-4} in SI units) on average.

Koenigsberger ratio

Many of the Koenigsberger ratios (Q_n) of the present samples give values of less than three. Since the Q_n of igneous rocks usually ranges from two to ten and is rarely below unity (NAGATA, 1961), the Q_n values of the present samples are rather small. This may be caused by a decay of NRM intensity associated with low temperature oxidation in a later stage. The existence of low temperature oxidation on the

samples is stated later.

Thermomagnetic analysis

Thermomagnetic analysis was made on 66 samples and a relationship between saturation magnetization and temperature (J_s-T curve) was obtained for each sample. The J_s-T curves are classified into the following three types (Fig. 4).

[Type I]: Forty-four samples show this type of J_s-T curve. The cooling curve is always slightly lower than the heating one. Saturation magnetization decreases smoothly with temperature, showing a sharp Curie point. Almost all the curves indicate a single phase with Curie temperatures, which range from 480 to 580° C in the heating stage and from 440 to 575° C in the cooling one. The Curie temperatures in the cooling stage are 20 to 30° C lower than those in the heating one. No substantial difference exists among the J_s-T curves of various rock types. Type I is one of the typical irreversible J_s-T curves. The irreversible character of this type may be due to a phase change of titanomaghemite during heating (KIKAWA, 1984). However, the relative amount of titanomaghemite seems to be small, considering the small difference between the heating and cooling curves.

[Type II]: Fifteen samples indicate this type of J_s-T curve. In Type II, a heating curve shows two slightly definable phases. Curie temperatures in the heating stage (480 to 555° C) are higher than those in the cooling one (440 to 550° C) by 5 to 30° C. After the thermal treatment, the saturation magnetization becomes slightly larger than it was originally.

[Type III]: Type III was obtained from eight samples. This type is characterized by almost reversible J_s-T curves. Curie temperatures range from 320 to 560° C in the heating stage and from 340 to 545° C in the cooling one.

All these thermomagnetic curves suggest that the magnetic career in the present

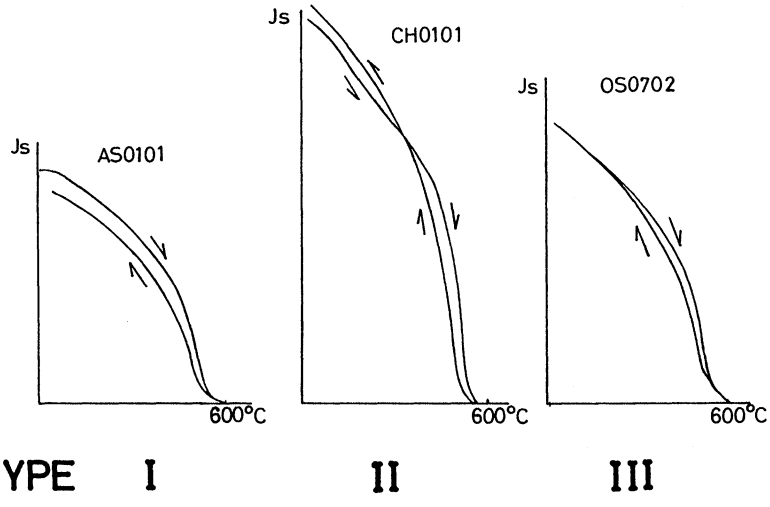


Fig. 4. Three typical types of thermomagnetic curves (further explanation, see text).

samples is titanomagnetite slightly suffered by low temperature oxidation. These results are confirmed by the microscopic observation stated below.

Opaque mineralogy

Polished thin sections of 69 samples were studied under reflected light. Opaque minerals in the samples are predominantly titanomagnetite, some of which had undergone low-grade low temperature oxidation. This is inferred from a light change in color, which is a transition from original reddish brown to resultant bright gray. The grain size of titanomagnetite is usually very small, between one micron and tens of microns, rarely up to 200 microns. Other observed magnetic minerals are (hemo)ilmenite mantled by titanomagnetite, ilmenite lamellae in titanomagnetite grains, (ilmeno)hematite and a small amount of iron sulfides.

4. Paleomagnetism

4.1 Paleomagnetic direction

The mean field direction and its confidence limit in each sampling site are summarized in Table 1. The majority of 95% confidence limits (α_{95}) are confined within 9°. The mean field direction from each volcano is shown in Table 2 and Fig. 5. No tilting correction was made on these results. We discuss the sense and amount of deformation of each volcano using these in-situ mean field directions (Section 5).

Result from Ashitaka Volcano

On the basis of the tephrochronology by YUI and FUJII (1988), sampling horizons of the present study cover the interval of at least several tens of thousand years. This suggests that the directional fluctuations caused by geomagnetic secular

Table 2. Summary of paleomagnetic results for each volcano after alternating field demagnetization. N is number of sampling sites; n is number of specimens; Jn is mean intensity of remanent magnetization ($\times 10^{-3}$ emu/g); DEC and INC are mean declination and inclination of remanence direction; K is precision parameter; α_{95} is radius of 95% confidence limit in degrees.

Volcano	N	n	Jn	DEC	INC	K	α_{95}
Ashitaka	19	119	0.7	355.8	+48.9	17.7	8.2
Fudeshima	21	92	1.5	2.4	+51.9	24.1	6.6
Okata	7	35	2.4	3.8	+45.8	14.5	16.4
Gyojanoiwaya	1	12	0.9	350.1	+47.7		
Izu-Oshima (total)	29	139	1.6	2.3	+50.3	21.8	5.9
Daruma (normal)	25	174	0.7	12.0	+50.7	19.3	6.8
Daruma (reversed)	22	193	0.5	189.1	−49.5	18.6	7.4
Daruma (total)	47	367	0.6	10.6	+50.1	19.3	4.9
Ida	10	69	0.9	10.9	+46.7	18.9	11.4
Tanaba	8	53	0.4	189.9	−47.7	17.5	13.7
Chokuro	7	45	0.4	352.2	+39.5	29.0	11.4
Jaishi	9	60	0.9	154.9	−58.0	17.1	12.8
Nanzaki	5	30	1.5	348.4	+36.9	115.1	7.2

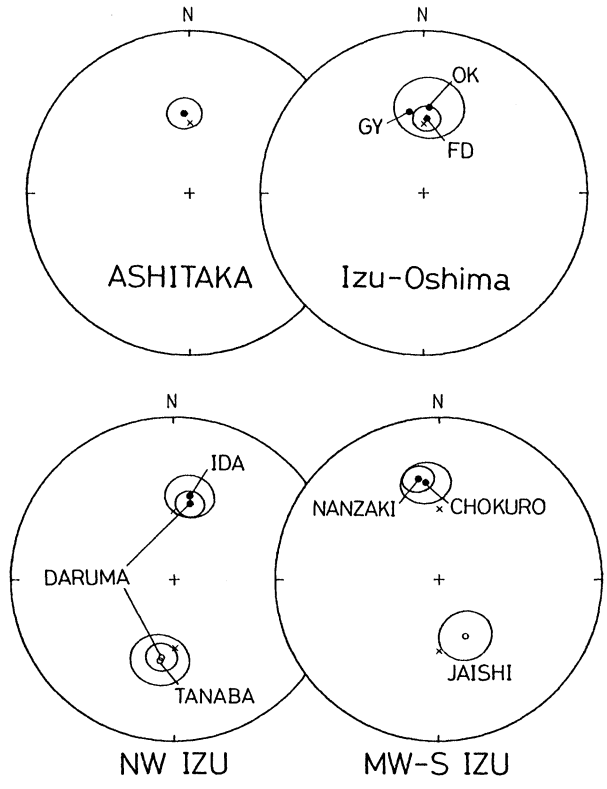


Fig. 5. Mean field direction and 95% confidence limit from each volcano. Abbreviations are OK, Okata Volcano; FD, Fudeshima Volcano; GY, Gyojanoiwaya Volcano; NW IZU, northwestern Izu Peninsula; and MW-S IZU, middle western to southern Izu Peninsula. Solid and open circles are on lower and upper hemispheres, respectively. ×: direction of the present axial geocentric dipole field in the study area.

variation is mainly averaged out, and that the mean result is regarded as a paleomagnetic direction.

The mean field direction of Ashitaka Volcano is aligned with the present axial geocentric dipole field ($D=0°$, $I=+54.3°$) within the confidence limit (Fig. 5, Table 2). This result is concordant with that by TSUNAKAWA and HAMANO (1988), which examined the paleomagnetism of the dykes in the upper part of Ashitaka Volcano.

Results from Izu-Oshima island

On Izu-Oshima island, the time interval covered by the sampling horizons are unclear due to the lack of sufficient chronologic data. However, the mean direction for the whole Izu-Oshima island seems to give a reliable paleomagnetic direction because each of the 29 sampling sites represents a different eruptive unit over the duration of three different volcanic activities. Moreover, the result from the dyke

swarm of Fudeshima Volcano shows an angular standard deviation of VGP (virtual geomagnetic pole) of 15.1° (+4.2°, −2.7°), which is within the range of the worldwide paleosecular variation dispersions at similar latitudes during the Brunhes epoch (MCELHINNY and MERRILL, 1975). This suggests that the samples of Fudeshima Volcano cover a period long enough to average geomagnetic secular variation.

Both the mean directions of Fudeshima and Okata Volcanoes agree with that of the present axial geocentric dipole field within the confidence limits (Fig. 5, Table 2). The direction of one site of Gyojanoiwaya Volcano shows no significant deflection from the results of the other two volcanoes. Consequently, the mean direction for the whole Izu-Oshima island is aligned with the present axial geocentric dipole field within the confidence limit (Table 2).

Results from the Izu Peninsula

According to the K–Ar ages (Section 2), the volcanism of Daruma Volcano seems to have been active for at least 0.2 million years. The samples from the 47 sites of Daruma Volcano have both normal and reversed polarities, which show an antipodal distribution (Fig. 5). These suggest that the time interval covered by the sampling sites of Daruma Volcano is long enough to average geomagnetic secular variation. Furthermore, the antipodal distribution implies that the contribution of stable secondary components is negligibly small. Mean directions for the other volcanoes are less reliable because of the small number of sampling sites (ten or less) and single polarity for each volcano (Fig. 5).

The mean directions in the Izu Peninsula consistently show some deflections from the direction of the present axial geocentric dipole field. In the northwestern Izu Peninsula, the mean directions consistently show clockwise deflections of an average of 10° from that of the axial dipole field (Fig. 5, Table 2), although the deflections of Ida and Tanaba Volcanoes are within the confidence limits. On the other hand, in the middle western to southern Izu Peninsula, the mean directions consistently show counterclockwise deflections of 8° to 25°. Moreover, all the mean inclinations from the Izu Peninsula seem to be slightly lower than that of the present axial geocentric dipole field, except for Jaishi Volcano, where the mean inclination nearly agrees with that of the axial dipole field (Fig. 5).

4.2 Magnetostratigraphy

Magnetostratigraphic correlation of each volcano is summarized in Fig. 6. All the samples of Ashitaka, Ida, Chokuro, Nanzaki, Okata, Fudeshima, and Gyojanoiwaya volcanoes have normal polarity, and all of Tanaba and Jaishi Volcanoes show reversed polarity. The early and late lava flows of Daruma Volcano have both normal and reversed polarities. Although these magnetic polarities are of in situ directions of remanence, they can be regarded as paleomagnetic polarities. This is because the studied volcanoes lie with gentle dips (Section 1). The gentle structure is confirmed by the fact that the mean field direction of each volcano has no large deviation (at most 20°) from that of the present axial geocentric dipole field (Fig. 5).

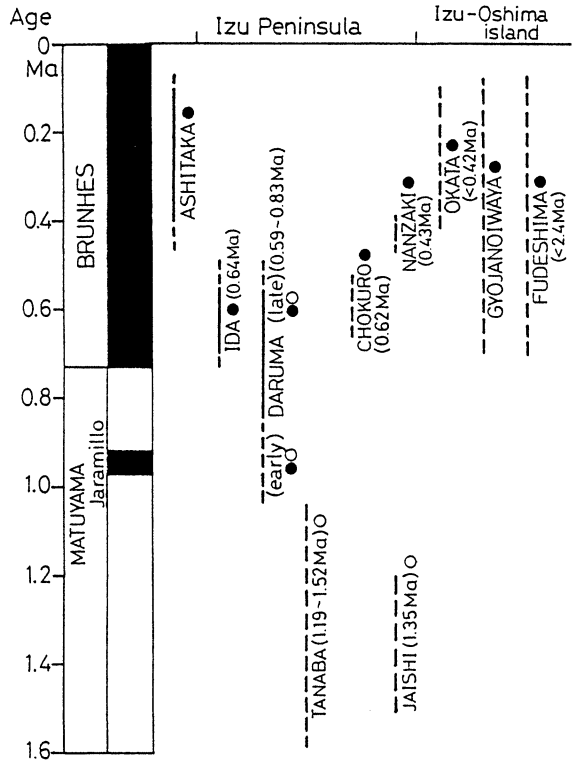


Fig. 6. Magnetostratigraphic correlation of each volcano. Magnetic polarities are shown with a solid circle (normal polarity) and an open circle (reversed polarity) for each volcano. K–Ar ages reported so far (see Section 2) are also shown. The magnetic polarity time scale is after MANKINEN and DALRYMPLE (1979).

On the basis of comparison of radiometric ages with magnetic polarities, the following correlations are drawn. The normal polarities of Ida, Chokuro, Nanzaki, and Okata Volcanoes are correlated to the Brunhes normal polarity chron because their K–Ar ages are younger than 0.7 Ma. The reversed polarities of Tanaba and Jaishi Volcanoes are correlated to the Matuyama reversed polarity chron because of their K–Ar ages. The normal and reversed polarities of the late lava flows of Daruma Volcano are correlated to the Brunhes normal and the Matuyama reversed polarity chrons, respectively, because their K–Ar ages range from 0.59 to 0.83 Ma. The normal polarity of the early lava flows of Daruma Volcano may be of the Jaramillo normal polarity subchron, because the early lava flows conformably underlie the late lava flows of Daruma Volcano (SHIRAO, 1981). According to the tephrochronology by YUI and FUJII (1988), the normal polarity of Ashitaka Volcano is correlated to the Brunhes chron. Since the ages of Fudeshima and Gyojanoiwaya volcanoes are estimated to be Quaternary (ISSHIKI, 1984), the

normal polarities of these volcanoes may be of the Brunhes chron. This is consistent with the older limit of radiometric age (2.4 Ma) of Fudeshima Volcano.

Figure 6 shows that the ages of the present samples range from 0.2 to 1.5 Ma. Since the Izu block is estimated to have begun to collide with Honshu between 0.8 and 1.5 Ma (Section 1), the present samples are appropriate for detecting the influence of the collision.

5. Tectonic Movement

The mean directions from both Ashitaka Volcano and Izu-Oshima island are aligned with the present axial geocentric dipole field (Section 4). Hence, we conclude that Ashitaka Volcano and Izu-Oshima island have undergone no post-cooling tectonic movements detectable by paleomagnetism.

As for the Izu Peninsula, we have another data set from the Quaternary volcanoes by previous paleomagnetic studies (Table 3 and Fig. 7). These results were obtained in the northeastern to middle eastern Izu Peninsula. The results from

Table 3. Paleomagnetic results from volcanoes in the northeastern to middle eastern Izu Peninsula by previous studies. N is number of sampling sites; DEC and INC are mean declination and inclination of remanence direction; α_{95} is radius of 95% confidence limit in degrees.

Unit	N	DEC	INC	α_{95}	Reference
Taga Volcano	5	340.7	+53.1	15.7	NAGATA *et al.* (1957)
Tenshi Volcano and accompanying Quaternary volcanics	12	345.6	+46.4	4.6	KOYAMA (1986)
Amagi Volcano	7	342.8	+46.9	5.0	KOYAMA (1986)

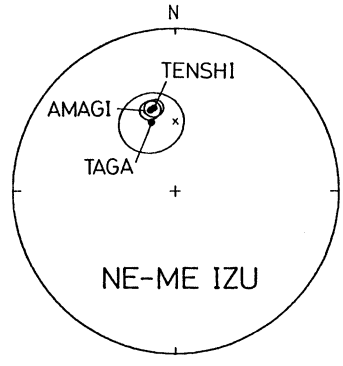


Fig. 7. Mean field directions from Quaternary volcanoes in the northeastern to middle eastern Izu Peninsula obtained by the previous works in Table 3.

Usami Volcano are not included in the present discussion because they are influenced by local deformations along a major fault (KOYAMA, 1981; Section 1). Figure 7 shows that the mean declinations consistently show counterclockwise deflections of 10° to 20°, and that the mean inclinations are slightly lower than that of the present axial geocentric dipole field. These observations in the northeastern to middle eastern Izu Peninsula are similar to the present results from the middle western to southern Izu Peninsula, which also show counterclockwise deflections of declination and slightly shallower inclinations (Fig. 5).

Figure 8 summarizes all the directions of declination obtained by this study and the previous works. These data suggest that the Izu Peninsula is divided into eastern and western halves, each of which consistently shows counterclockwise and clockwise deflections of declination, respectively. Such systematic deflections of paleomagnetic directions suggest that they were caused by some regional tectonic movement. If this is the case, the directional difference between the eastern and

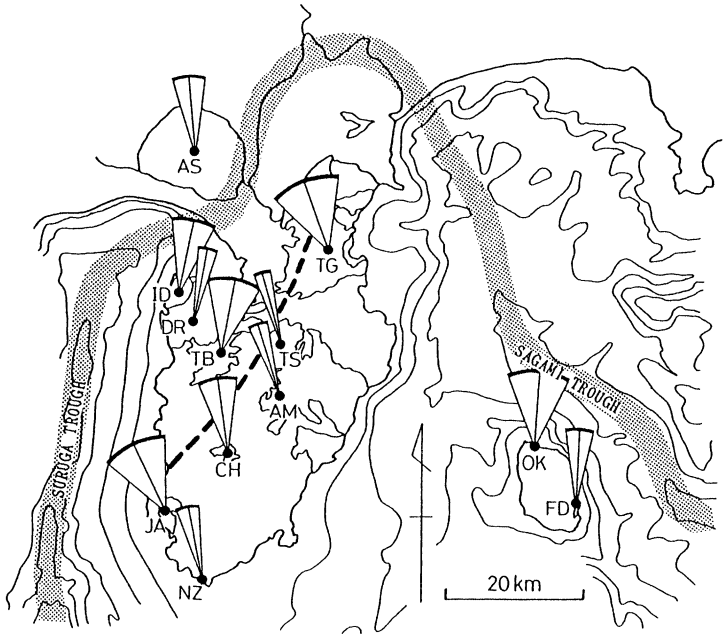


Fig. 8. Mean direction of paleomagnetic declination and its confidence limit for each volcano. AS (Ashitaka Volcano), ID (Ida Volcano), DR (Daruma Volcano), TB (Tanaba Volcano), CH (Chokuro Volcano), JA (Jaishi Volcano), NZ (Nanzaki Volcano), OK (Okata Volcano), and FD (Fudeshima Volcano) are obtained by this study. TG (Taga Volcano), TS (Tenshi Volcano), and AM (Amagi Volcano) are by the previous works in Table 3. All the directions with reversed polarity were converted to those with normal polarity. The locality of each symbol is approximately on the center of the sampling sites in each volcano. Broken line shows the approximate boundary of tectonic province in the Izu Peninsula (see text). Shaded belt: material boundary between plates (NAKAMURA *et al.*, 1984).

western halves means that each half of the Izu Peninsula has been controlled by a different way of tectonic deformation in the Quaternary time. The approximate trace of the boundary between the eastern and western halves is shown in Fig. 8.

Many geological and geophysical data also suggest the existence of this boundary and the two tectonic provinces bounded by it. In the light of stratigraphic survey (KOYAMA, 1986), this boundary nearly agrees with a major fault-concentrated belt formed in the early Quaternary age. According to active fault maps (MURAI and KANEKO, 1974; HOSHINO *et al.*, 1978), normal faults are dominant in the western half of the Izu Peninsula, while right-lateral strike-slip faults are dominant in the eastern half. On the basis of in-situ stress measurement, focal mechanisms of shallow earthquakes, and geological evidence, N-S directions of maximum horizontal compressive stress are dominant in the western half, while NW-SE directions are dominant in the eastern half (TSUKAHARA and IKEDA, 1983, 1987; HOSHINO, 1984). In addition, on the basis of strikes of dykes, NAKANO *et al.* (1980) proposed a similar difference of stress field defined by this boundary.

What causes the deflections of paleomagnetic directions and the tectonic provinces? There are many different modes of tectonic deformation explaining the deflection of a paleomagnetic direction. Here let us try to consider the two simplest modes of deformation; they are (A) tilting around a horizontal axis and (B) rotation around a vertical axis. In many cases, tilting causes a deflection in both inclination and declination of a paleomagnetic direction, whereas rotation causes a deflection only in declination. If observed paleomagnetic directions show significant shallower inclinations than that of the present, mode A is preferable for explaining the observed results. In Tables 2 and 3, most of the volcanoes consistently show shallower inclinations than that of the present axial geocentric dipole field, although they are within the confidence limits except for four volcanoes (Chokuro, Nanzaki, Tenshi, and Amagi Volcanoes). Thus, we prefer mode A to explain the observed deflections of paleomagnetic directions. We cannot, however, completely deny the possibility of deformation including tectonic rotation, such as an intermediate type between modes A and B.

Mode A attributes the deflections of paleomagnetic directions to tilting around a horizontal axis. Figure 9 shows the tilting direction and angle for each volcano calculated from the paleomagnetic results on the basis of this mode. In this calculation, it is assumed that the original paleomagnetic vectors are parallel to the present axial geocentric dipole field. The calculated results show that the western half of the Izu Peninsula has tilted westward to southwestward, whereas the eastern half has tilted eastward to southeastward (Fig. 9).

In the northeastern and northwestern Izu Peninsula, some geological and geodetical data support mode A. KUNO (1972) suggested that Yugawara and Taga Volcanoes in the northeastern Izu Peninsula (Fig. 2) have tilted southeastward on the basis of the level change of some key horizons. According to the geologic map of Ida and Daruma Volcanoes by SHIRAO (1981), many of the lava flows around the present sampling sites show northwestward to southwestward dips of 10° to 20°. These observations are consistent with mode A. SUZUKI *et al.* (1975) analyzed the

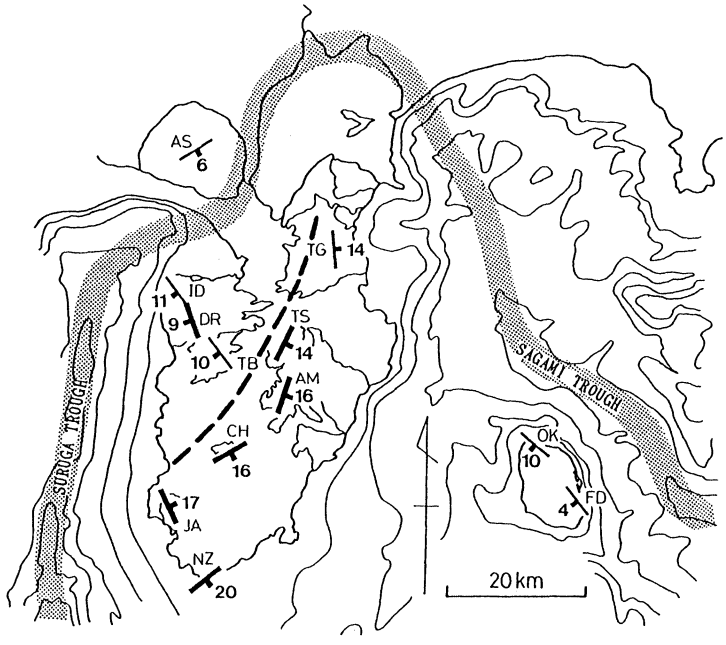


Fig. 9. Tilting direction and angle for each volcano calculated from the paleomagnetic direction on the basis of deformation mode A (see text). The reliability of each data source is shown with symbols: thick symbol, the data of which source paleomagneic direction is significantly deflected from that of the present axial geocentric dipole field; thin symbol, the data of which source direction agrees with that of the axial dipole field within the confidence limit. Other details are the same as those in Fig. 8.

data of precise levelling during the interval from 1903 to 1967, and revealed that the western coast of the Izu Peninsula has tilted westward to southwestward. This is also similar to the present result based on mode A (Fig. 9).

On the basis of the present paleomagnetic results and the other geological and geophysical evidence stated above, we propose a unified model of the Quaternary tectonism in the study area (Fig. 10). The explanation of the model is stated as follows. The Izu block is divided into western and eastern halves (W-IZU and E-IZU in Fig. 10). The W-IZU has tilted westward to southwestward, while the E-IZU has tilted eastward to southeastward. The W-IZU is mechanically decoupled with the E-IZU by the central boundary. In the W-IZU, tensional stress in an E-W direction and active normal faults are dominant because of slub pull of the subducting plate along the Suruga trough. This may be an analogue of the tectonism on seaward slopes of trenches, where normal tensional events commonly accompany bending of lithosphere prior to its subduction (CHAPPLE and FORSYTH, 1979). The westward or southwestward tilting of the W-IZU may also be a part of such lithosphere bending. In the E-IZU, compressional stress in a NW-SE direction and resultant active strike-slip faults are dominant. These compressional features and the

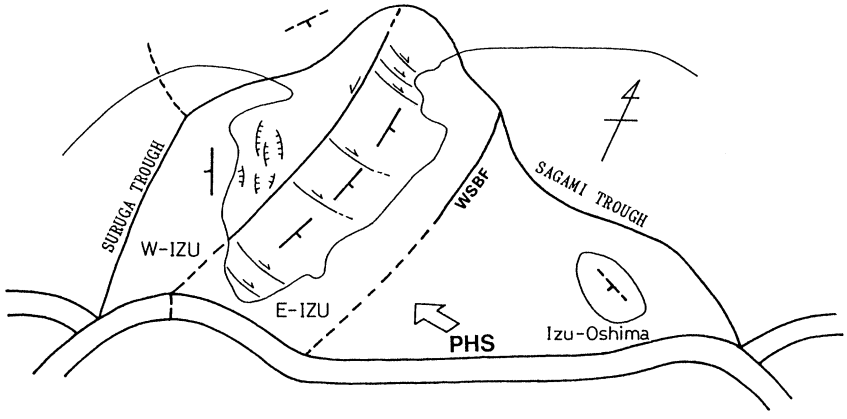


Fig. 10. Schematic diagram of deformations in the Izu Peninsula and adjacent areas (see text). Approximate tilting direction (Fig. 9) and major active faults are also schematically shown. Faults with arrows and fringes indicate strike-slip and normal faults, respectively. Abbreviations are W-IZU and E-IZU, the western and the eastern halves of the Izu Peninsula (see text); WSBF, the West-Sagami-Bay fault after ISHIBASHI (1976, 1980); PHS, relative convergence direction of the Philippine Sea plate against Honshu.

eastward or southeastward tilting of the E-IZU may be a manifestation of the deformation associated with the collision of the Izu block against Honshu.

On the basis of focal mechanisms of shallow earthquakes, active faults, geodetic data, and geomorphology, NAKAMURA and SHIMAZAKI (1981) and NAKAMURA *et al.* (1984) proposed a fan-shaped alignment of axes of maximum horizontal tensional stress (Fig. 11), and attributed them to anticlinal bending of the northern tip of the Philippine Sea plate including the Izu block in the late Quaternary time. This bending hypothesis predicts anticlinal tilting of the Izu block, where the axis of the anticline passes the central part of the Izu Peninsula and plunges north-northwestward (Fig. 11). Such bending is expected to tilt the northeastern part of the Izu Peninsula northeastward and the western part westward, respectively (Fig. 11). This tilting pattern is similar to that of our model in the W-IZU and in the northern part of the E-IZU (Figs. 10 and 11). The E-W direction of horizontal tensional stress and the normal faults of N-S strikes in the W-IZU stated above are also consistent with the bending hypothesis. In the southern part of the E-IZU, the southeastward direction of tilting expected from our model (Figs. 9 and 10) is, however, different than the result expected from the bending hypothesis (Fig. 11). Moreover, the bending hypothesis does not sufficiently explain the compressional features in the E-IZU, such as the dominance of active strike-slip faults. Some modification seems to be needed for the bending hypothesis to minimize such discrepancy.

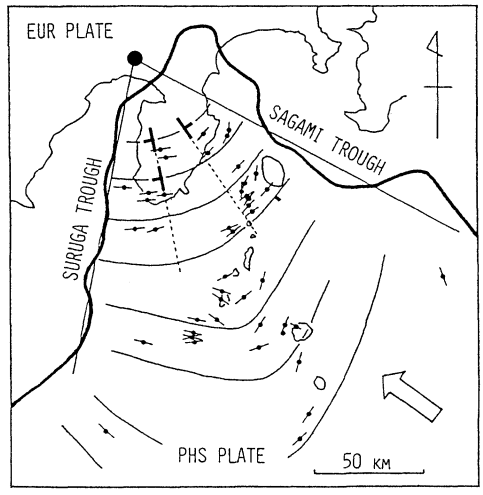


Fig. 11. Fan-shaped trajectories of tensional axes obtained from focal mechanism solutions of shallow earthquakes in the northern tip of the Philippine Sea plate (after NAKAMURA *et al.*, 1984). Large open arrow indicates the direction of convergence between the Philippine Sea plate and Honshu. Tilting directions expected from the bending hypothesis of NAKAMURA *et al.* (1984) (see text) are added by the authors.

6. Concluding Remarks

The paleomagnetic directions obtained from both Ashitaka Volcano and Izu-Oshima island are nearly aligned with the present axial geocentric dipole field. This means that these areas have undergone no tectonic deformation detectable by paleomagnetism. Two anomalous paleomagnetic directions exist in the Izu Peninsula, one of which is on average 10° clockwise deflections in the northwestern part and the other is 8° to 25° counterclockwise deflections in the northeastern to southern parts. These deflections can be explained by differential tilting of the eastern and the western halves of the Izu Peninsula in the Quaternary time.

TONOUCHI and KOBAYASHI (1983) proposed that a subduction zone between the Izu-Bonin arc and Honshu discontinuously jumped seaward from the Hayama-Mineoka and Setogawa belts to the Sagami and Suruga troughs and is jumping to the West-Sagami-Bay fault (ISHIBASHI, 1976, 1980) and its extension. If this is the case, the Izu block is now being accreted to central Honshu. The Tanzawa block, which is located to the north of the Izu block and has a heavily deformed structure,is regarded as an allochthonous terrane accreted prior to the collision of the Izu block (e.g., NIITSUMA and MATSUDA, 1985). The geologic structure of the Tanzawa block is, however, not sufficiently understood as a record of post-collisionl deformations. The Quaternary tectonism of the Izu block studied here may be an example of an initial phase of a post-collisional deformation and may also be a key for understand-

ing past accretion processes.

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